

Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25EVT101 – ELECTRIC AND HYBRID VEHICLES

(For EVT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) List and explain the major EV technologies, including body design, propulsion system, energy sources, energy management, and overall system optimization.	7	L2	1	3,4
b) Compare Electric Vehicles (EVs) and Internal Combustion Engine (ICE) vehicles with respect to efficiency, emissions, maintenance requirements, and performance characteristics.	7	L2	1	3,4
c) List the major barriers affecting the development, large-scale deployment, and adoption of Electric Vehicles.	6	L2	1	3,4
2. a) Draw the general layout of an Electric Vehicle (EV) and explain the roles of propulsion system, energy source, power electronics, and auxiliary systems in EV operation.	8	L2	1	3,4
b) Compare single-motor and multi-motor drive configurations in EVs with neat sketches, highlighting their advantages and limitations.	4	L2	1	3,4
c) Draw and explain the torque-speed characteristics of an Electric Vehicle. Define speed ratio and explain how fixed-gear and variable-gear arrangements influence EV performance at different speed ratios.	8	L2	1	3,4

Unit – II

3. a) Interpret the In-Wheel Motor configuration used in EVs. With a neat diagram, discuss its working principle.	7	L2	2	3,4
b) Draw the functional block diagram of an Electric Propulsion System. Discuss the key factors that influence the selection of a suitable propulsion system for EVs.	7	L2	2	3,4
c) Compare fixed-gear and variable-gear transmissions in EVs in terms of motor rating, inverter rating, cost, size, weight, efficiency, and overall reliability.	6	L4	2	3,4
4. a) Analyze the operation of a Four-Quadrant DC Chopper used in EV drives. With neat diagrams, describe switch conduction in all four quadrants.	7	L4	2	3,4
b) Draw the circuit diagram of a Three-Phase Full-Bridge Inverter and explain its operation under the 120° conduction mode. Identify the conduction of switches in each interval.	7	L3	2	3,4
c) Evaluate the advantages and limitations of soft-switching power converters compared to hard-switching converters in EV applications.	6	L2	2	3,4

Unit – III

5. a) Illustrate the torque-speed characteristics of a DC motor and their relevance to EV propulsion.	7	L2	3	3,4
b) Make use of a Two-Quadrant Zero-Voltage Transition (ZVT) converter-fed DC motor drive, to explain principle of operation and major modes.	7	L3	3	3,4



c)	Outline the various speed-control methods used for DC motors in EV systems.	6	L2	3	3,4
6. a)	Draw single phase ARS inverter and identify various modes of operation.	8	L3	3	3,4
b)	Explain the Field-Oriented Control (FOC) method used for speed control of induction motor drives in EVs.	8	L3	3	3,4
c)	Outline the concept of Equal-Area PWM and list out its advantages.	4	L2	3	3,4

Unit – IV

7. a)	With neat diagrams, analyze the power-flow paths and operational modes of a Series Hybrid Electric Vehicle (HEV).	8	L3	4	3
b)	Demonstrate torque-coupling and speed-coupling mechanisms used in parallel hybrid electric vehicles.	8	L2	4	3
c)	Illustrate the schematic diagram of a parallel hybrid electric vehicle.	4	L2	4	3
8. a)	Classify the different types of HEVs and draw layout diagrams for each architecture.	8	L2	4	3
b)	Illustrate the power flow and operation modes of a Parallel HEV.	8	L2	4	3
c)	Outline the advantages of Hybrid Electric Vehicles (HEVs) over pure Electric Vehicles (EVs).	4	L2	4	3

Unit – V

9. a)	Analyze how soft-switching improves efficiency and reduces switching losses in EV wireless charging systems.	7	L4	5	3
b)	Summarize the charging and discharging characteristics of a Lithium-Ion battery. Discuss why these characteristics are important for EV operation and battery life.	7	L2	5	3
c)	Outline domestic and public EV charging infrastructures. Discuss the challenges associated with each type.	6	L2	5	3
10. a)	List the major challenges in EV charging infrastructure, including grid integration, power demand, and station placement.	7	L4	5	3
b)	Compare the charging behavior of a lithium-ion battery with ultra capacitor.	7	L2	5	3
c)	Outline basic principle of inductive charging. Explain how magnetic coupling enables wireless power transfer in EV applications.	6	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
